

Modelling Room Acoustics with a GPU

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Abstract

This is the bit where you summarise what is in your thesis.

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This template has been produced with help from many former students who have shown different ways of doing things. Please make suggestions for further improvements.

Chapter 1

Introduction

“There is no competition of sounds between a nightingale and a violin.”

Dejan Stojanovic

The way we perceive sound depends on the environment in which we hear it as well as how the sound was created. Listening to a sound in a space like a church or a cathedral is very different to listening in a bedroom. Being able to predict how a space will affect a sound has practical use in the design of new spaces, particularly those spaces where acoustic performance is critical. A great deal of effort goes into the design of new concert halls. This would be greatly enhanced by accurate acoustic models of spaces. The aim then, of research in this area, is to provide an accurate prediction of how a space will sound after being given its physical properties.

The key technique currently used for these simulations is the finite-difference time-domain (FDTD) method for finding approximate solutions to the wave equation across a three dimensional space. This forms a simple stencil. This stencil process across a vast number of points in memory is well suited to the use of graphics processing units (GPUs) which have high band width graphics memory and can operate on very many points in a massively parallel fashion.

In the work leading up to this report, a novel technique was devised and implemented that performs these simulations. This approach was designed to make best use of graphics memory, which is relatively scarce.

Chapter 2

Background and Literature Review

“He who loves practice without theory is like the sailor who boards ship without a rudder and compass and never knows where he may cast.”

Leonardo da Vinci

2.1 Room Acoustics

2.1.1 Classical Acoustics

Sound is modelled using the wave equation:

$$\frac{\partial^2 P}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 P}{\partial t^2} = 0 \quad (2.1)$$

where scalar field P is the deviation from the ambient air pressure, and c is the speed of sound. This result is derived by, among others, Richard Feynman in [?]. It is trivial to generalise this for 3 dimensional systems, which yields

$$\nabla^2 P - \frac{1}{c^2} \frac{\partial^2 P}{\partial t^2} = 0 \quad (2.2)$$

where

$$\nabla \equiv \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z} \quad (2.3)$$

Intuitively we know that a room’s acoustic properties are dependent on its size and what materials form the walls and other surfaces. This is reflected in the solutions to the wave equation, where almost all environmental data is encoded in boundary conditions

imposed on the P scalar field. In real world applications these boundary conditions are too sophisticated for the wave equation to be solved analytically. This sophistication stems from both the shape of the room and the reflective properties of its boundaries. These reflective properties are a function of the frequency of the sound reflected.

2.1.2 Finite Difference Schemes

Finite difference schemes are therefore used to solve the wave equation for complex room shapes. These can provide an approximate numerical solution to the continuous wave equation.

A simple equivalence between first and second order derivatives and their corresponding central difference schemes is shown in 2.4 and 2.5.

$$\frac{df(u)}{du} \approx \frac{f(u+h) - f(u-h)}{2h} \quad (2.4)$$

$$\frac{d^2f(u)}{du^2} \approx \frac{f(u+h) - 2f(u) + f(u-h)}{h^2} \quad (2.5)$$

This can be easily discretised if h is set to be the distance in space or time between samples.

When discretising the space within the room, the scalar field of pressure within room is represented as a three dimensional array of floating point numbers. The letters l , m , and p are chosen to represent the samples index in each spacial dimension within the array, with n giving the time step. This means $P_{l,m,p}^n$ gives an approximation to P at the point (x, y, z) at time t where $x = lH$, $y = mH$, $z = pH$ and $t = nT$. T and H are the sample rates in time and space respectively.

When equation 2.2 is expanded out, it is shown that in order to compute a numerical approximation of the wave equation, discrete equivalents are required to the derivatives in time, and in each of the three spacial dimensions.

$$\frac{\partial^2 P}{\partial t^2} = c^2 \left(\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + \frac{\partial^2 P}{\partial z^2} \right) \quad (2.6)$$

To replace these continuous partial derivatives a new finite difference operator, δ , is introduced. From 2.5 the following relations can be derived by simply replacing h with H :

$$\delta_t^2 P_{l,m,p}^n \equiv \frac{1}{T^2} \left(P_{l,m,p}^{n+1} - 2P_{l,m,p}^n + P_{l,m,p}^{n-1} \right) \approx \frac{\partial^2 P}{\partial t^2} \quad (2.7)$$

$$\delta_x^2 P_{l,m,p}^n \equiv \frac{1}{H^2} \left(P_{l+1,m,p}^n - 2P_{l,m,p}^n + P_{l-1,m,p}^n \right) \approx \frac{\partial^2 P}{\partial x^2} \quad (2.8)$$

$$\delta_y^2 P_{l,m,p}^n \equiv \frac{1}{H^2} \left(P_{l,m+1,p}^n - 2P_{l,m,p}^n + P_{l,m-1,p}^n \right) \approx \frac{\partial^2 P}{\partial y^2} \quad (2.9)$$

$$\delta_z^2 P_{l,m,p}^n \equiv \frac{1}{H^2} \left(P_{l,m,p+1}^n - 2P_{l,m,p}^n + P_{l,m,p-1}^n \right) \approx \frac{\partial^2 P}{\partial z^2} \quad (2.10)$$

It is now possible to use these operators to discretise 2.6.

$$\delta_t^2 P_{l,m,p}^n = c^2 \left(\delta_x^2 P_{l,m,p}^n + \delta_y^2 P_{l,m,p}^n + \delta_z^2 P_{l,m,p}^n \right) \quad (2.11)$$

Equation 2.11 can then be expanded using the results of equations 2.7-2.10. This yields

$$P_{l,m,p}^{n+1} - 2P_{l,m,p}^n + P_{l,m,p}^{n-1} = \frac{c^2 T^2}{H^2} \left(P_{l+1,m,p}^n + P_{l-1,m,p}^n + P_{l,m+1,p}^n + P_{l,m-1,p}^n + P_{l,m,p+1}^n + P_{l,m,p-1}^n - 6P_{l,m,p}^n \right) \quad (2.12)$$

Letting the Courant number be defined as

$$\lambda \equiv \frac{cT}{H} \quad (2.13)$$

and our stencil

$$S = P_{l+1,m,p}^n + P_{l-1,m,p}^n + P_{l,m+1,p}^n + P_{l,m-1,p}^n + P_{l,m,p+1}^n + P_{l,m,p-1}^n \quad (2.14)$$

gives a somewhat neater version of equation 2.12

$$P_{l,m,p}^{n+1} = (2 - 6\lambda^2)P_{l,m,p}^n + \lambda^2 S - P_{l,m,p}^{n-1} \quad (2.15)$$

More than simply being neat, this is now a representation of how to calculate new values for the three dimensional array from the previous ones using a simple stencil operation.

2.2 Graphics Processing Units (GPUs)

2.3 The Proposed Data Structure and Method

2.4 Other Methods

While no work was done on this, I am going to talk about it anyway.

$$h(x) = f \otimes g = \int_{-\infty}^{\infty} f(x-u)g(u)du = \mathcal{F}^{-1}\left(\sqrt{2\pi}\mathcal{F}[f] \cdot \mathcal{F}[g]\right) \quad (2.16)$$

You can use a label on a figure to refer to it later. The university crest is in 5.1. Note that you should not use phrases like “the figure above” or “the following figure” since Latex may move the figure relative to the text if it cannot be fitted onto the current page.

Chapter 3

Design and Implementation

“The first 90 percent of the code accounts for the first 90 percent of the development time...The remaining 10 percent of the code accounts for the other 90 percent of the development time.”

Tom Cargill

3.1 Software Development Process

3.2 Debugging and Profiling Tools

Chapter 4

Results and Analysis

4.1 Some results

Here are some results.

4.1.1 More results

When showing results you are likely to use tables and graphs. You can create tables easily in Latex.

File names	Satellite	Resolution
worldr	Meteosat	5km
worldg	Meteosat	5km
worldb	Meteosat	5km

Table 4.1: This is a simple table. More complicated tables can have headings which pass over more than one column

Chapter 5

Conclusions

5.1 Future Work



Figure 5.1: The University Crest

Appendix A

Stuff which is too detailed

Appendices should contain all the material which is considered too detailed to be included in the main bod but which is, nevertheless, important enough to be included in the thesis.

Appendix B

Stuff which no-one will read

Some people include in their thesis a lot of detail, particularly computer code, which no-one will ever read. You should be careful that anything like this you include should contain some element of uniqueness which justifies its inclusion.

Bibliography

- [1] L.Lamport. *1986 Latex User's Guide and Reference Manual*. Addison Wesley.
pp242.
- [2] F.Bloggs. *1993 Latex Users do it in Environments* Int. Journal of Silly Findings.
pp 23-29.